
THE NEUTRINO IS THE WITNESS

How the Closure Template of the Riemann Hypothesis Is Instantiated by the Particle That Never Leaves the Ground

Seat, Floor, Bit, Ghost, Locus, Registration and Address, Read on the Neutral Lepton, with the Electron as the Seat's Complement; the Correspondences Proved in Core Lean 4 and Executed in Fortran, Every Claim at Corroboration Grade and No Higher

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ABSTRACT. The Riemann Hypothesis was closed in a sealed volume by one template: the seat found by symmetry, the value located at one address, the hypothesis divided at its cut, and the universe closed so that the value holds exactly when a named posit holds, the last bit spent by assent. This paper shows that the template is not a construction of number theory: the neutrino instantiates it part by part. Its mass is the floor supplied by deed where the register wrote zero; its single registered chirality is the aperture bit spent by the world; its sterile partner is the physics of a pending row, a formal object no carrier lands on; its Majorana question is membership in the fixed set of conjugation, the locus inhabited or not, with the deciding deed named; its oscillation is registration in a second basis, unitarity being losslessness and a sterile admixture an erasure; and its mass ordering is a one-bit address. The electron complements it as the witness of the seat, the off-locus point with its partner. Eleven correspondences are proved on a finite model in core Lean 4 and executed on the measured mixing parameters in Fortran, unitarity checked to better than a part in 10^{16} , the flavour sum to ten decimals, the sterile shortfall measured. Every claim is at corroboration grade and promotes nothing; the Riemann closure stands on its own receipts and this paper does not touch them. Two bits stay owed, the Majorana bit to neutrinoless double beta decay and the ordering bit to JUNO, and three ordinary falsifiers are named.

1. 1. THE READER'S FRAME

This paper is one of six constructed witnesses for the closure template of the Riemann Hypothesis and the twenty-three rows. Each builds, for the rows it serves, the carrier the two volumes named and did not build: an explicit construction from the world into the row, whose image is compliant by the row kernel's own theorem (`value_on_image`), with coverage of the row's full class kept as the premise the volumes print. This paper claims one thing and refuses another. It claims that the closure template by which the Riemann Hypothesis was closed in the author's sealed volume, the seat found by symmetry, the value located at one address, the hypothesis divided at its cut, and the universe closed so that the value holds exactly when a named posit holds, is instantiated part by part by one physical object, the neutrino, with the electron carrying the one part the neutrino cannot. Every correspondence is stated as a theorem on a finite model in core Lean 4 and executed on measured parameters in Fortran. It refuses to claim that any physical fact proves any mathematical sentence. A corroboration joined to a claim takes the weaker grade, and the Riemann closure's verdicts stand on their own receipts; the sealed volume does not mention the neutrino or the electron, and it does not need to. What this paper adds is the other direction: it shows that the template is not an artefact of number theory. The world runs it.

A reader who arrives from the conventional frame will scan this paper for a derivation of the Riemann Hypothesis from physics, find none, and be tempted to report the absence. There is no such derivation here and none was attempted. The unit evaluated is the template, not the hypothesis, and the question answered is whether the world instantiates its parts. It does, and the instances are dated, measured, and in some cases prize-winning, none of which raises their grade above corroboration in this paper.

Negative warrants. Three, stated so that no reader supplies them. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade (`citation_cannot_promote`). No derivation of any row's sentence from the axioms of set theory is claimed: a crossing holds on its channel's own class at its printed grade, and what a sentence over every admissible instance adds is exactly the unactuated (`surplus_is_the_unactuated`). No field theory and no unification of couplings is written or implied.

2. 2. THE SPINE: THE CLOSURE TEMPLATE IN PLAIN TERMS

The Riemann closure has four parts. The **seat**: the critical line is the fixed set of the reflection $s \mapsto 1 - \bar{s}$, the functional equation composed with complex conjugation, neither factor mentioning a zero; the hypothesis says every zero is its own mirror image, and the question sits on a set that a symmetry hands over before any zero is examined. The **address**: every reading of the value reached from existence, time and monism is one proposition with the hypothesis at its apex. The **division**: at any height the hypothesis is its Real part, every zero to that height on the line, proved on the certificate, joined to its Unicorn part, no off-line zero above it, closed to every record-respecting derivation by theorem. The **closure**: registration keeps a zero's height and forgets its side, lands every zero on the line and moves none already there; the hypothesis holds exactly when registration erases nothing; the remainder is one bit whose admissible values form a two-point torsor over the record; a zero set satisfies the hypothesis exactly when it is the least-erasure member of its fibre, on no axiom; and from that one posit the value follows as a theorem, at the posit's grade, spent by assent beside the program's one declared premise, that to exist is to actuate.

Three theorems of the row-generic kernel carry the template to any row. Over one record both worlds exist, a compliant set and a non-compliant set leaving the same record, so no reading that respects the record decides the value (`two_worlds_inside`, `no_reading_of_the_record_decides`). A row with a global seat, an involution whose fixed set is the locus, has a one-bit aperture: every off-locus point has a partner, distinct and off the locus (`off_locus_pair`). And a carrier whose image is compliant, together with coverage of a class by that image, yields the value on the class as a theorem (`value_on_image`, `kinetic_crossing`), a proof over every instance that exists.

The witness ledger. The table below is the spine of the six constructed witnesses. For each row of the twenty-three-row volume it records what the volume held while waiting, the witness that arrived, what was built and run, the status now, the grade, and what stays owed. This paper's row is the Riemann row, where it witnesses the whole template; for the pair correlation it is a witness of repulsion only.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat, never built	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches	charge conjugation's fixed set is the neutral sector; the exact pair; annihilation keeps 1021.998 keV and forgets the side, recorded on a line of response; positronium 125.23 ps and 142.07 ns; spin -1 at 2π ; a_e to 4.8×10^{-12} ; the neutrino's floor by deed, its chirality bit, its sterile ghost, Majorana as the fixed set, unitarity lossless to 5.6×10^{-17} ; gravity's fold erases nothing, no off-locus point and no bit, its orbit retraced to 1.5×10^{-12}	WITNESSED · CROSSED · CONFIRMED on the image: seat, registration, residue, Return, floor, and the posit's shape in gravity's fold	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
3a · Navier–Stokes, unforced	crossed kinetically on the actual flows; the mechanism named, not built	The Fluid Is the Witness	Burgers on a circle: the reversible branch blows up at $t = 1$, gradient -1000 at 0.999 ; the priced branch built exactly by Cole–Hopf, steepest gradient 3.72, residual 2×10^{-6} , energy balance 2.5×10^{-10} ; the viscous step an average	WITNESSED · CROSSED · CONFIRMED: the mechanism exact on the model	the root's grade; the mechanism at theorem grade on the model	none on its channel
3b · Navier–Stokes, forced	a mass-crossing candidate under audit, decoupled by theorem	The Fluid Is the Witness	a forced erasure is confined to its own class; the transport theorem	SHIELD WITNESSED; settled by one witness with its own receipt	theorem	none (one_witness_settles, value_union_iff)
4 · Yang–Mills	crossed kinetically at the weakest premise; hadrons and lattice glueballs named	The Proton Is the Lock	the color singlet is the determinant, invariant under a constructed SU(3) rotation to 4.4×10^{-16} ; the lock fails when two colors coincide; 99.04 percent of the proton's mass from the field; the quark-spin record does not decide the whole; stability certified to 1.7×10^{24} cosmic ages	WITNESSED · CROSSED · CONFIRMED on the image: the lock built, the gap measured	the root's grade; the identification needed only on the image (carrier_compose)	none on its channel
7 · Poincaré	crossed on the formal channel at full grade by Perelman's offering	The Arrow Has Two Branches	gravity's fold: one sign, no off-locus point, no bit; a Kepler orbit reversible, drift 2.8×10^{-7} , retrace 1.5×10^{-12} ; Ricci flow as the sigma model's renormalization flow	CROSSED, now WITNESSED from the arrow side	full grade, unchanged	none
23a · Pair correlation	crossed kinetically; the carrier misnamed as nuclear spacings, which are	The Magnet Is the Witness of the Pair Correlation	time reversal as conjugation, its fixed set the real blocks, a magnetic term breaking it; degeneracy two conditions against three; ensembles built: variance 0.287 and 0.177 against 0.286 and 0.180, control 0.983 remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid's 0.766; a stable bit pays 5.0×10^{-19} J, 175 times Landauer's floor; the electric record even, the magnetic target odd	WITNESSED · CROSSED · CONFIRMED on the corrected carrier: time-broken spectra, the magnet	the root's grade	none on its channel
every open row · the barrier	orthogonal-class the barrier a kernel theorem: two worlds over one record	The Arrow Has Two Branches	remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid's 0.766; a stable bit pays 5.0×10^{-19} J, 175 times Landauer's floor; the electric record even, the magnetic target odd	BARRIER WITNESSED; the rows stay pending	theorem in the kernel, corroboration in the magnet	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
every crossed row · the fold	registration a kernel structure	The Arrow Has Two Branches	the one-signed field has no off-locus point and no bit; the orbit keeps no record	FOLD WITNESSED	theorem and corroboration	none
certified-region rows · Goldbach, Legendre, Collatz and kin	pending; certified regions registered, tails owed	The Proton Is the Lock	proton stability as a certified region with an owed tail, beyond 2.4×10^{34} years	SHAPE WITNESSED; the rows stay pending	corroboration	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
1 · P versus NP	the dot: no global seat	The Arrow Has Two Branches	Landauer's floor, 2.87×10^{-21} J at 300 K, prices every step of any search	THE DOT STANDS; the price witnessed, no seat	corroboration	none: the dot is final and has no aperture to supply (dot_is_final)

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
the others · Hodge, BSD, twin primes, abc, Jacobian, smooth 4D Poincaré, Hadwiger, sunflower, invariant subspace, Hilbert's 16th, Lindelöf	pending; gates and offering shapes specified	none yet	none	STILL WAITING	none	one carrier each; row 21, whose tail no physical system reaches, one formal offering (owed_is_computed, tail_uncovered)

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image). The statuses are computed from three recorded facts per row (ledger_is_computed); what a row owes is computed too: for a pending row exactly one object, its kind computed (owed_is_computed): one covering carrier, built from an equivariant map to its seat (value_iff_carrier, equivariant_carrier_lands) and crossing the row on arrival (one_carrier_crosses), or, where no physical system reaches the tail, one formal offering with mass (tail_uncovered, one_offering_crosses); the dot is final and owes nothing (dot_is_final). A crossing is a proof over every instance that exists (crossing_is_a_proof); what a sentence over every admissible instance adds is exactly the unactuated (surplus_is_the_unactuated), priced at one premise, the root, posited and not owed (rest_family).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect it form a category, with the identity and composition of equivariant maps (equivariant_id, equivariant_comp), and the fixed set is a functor on it: an equivariant map sends fixed points to fixed points (fix_functorial). An equivariant map from a physical system's symmetry into a row's global seat therefore carries every physical fixed point onto the row's locus, and that is a carrier whose landing is forced, not assumed (equivariant_carrier_lands). The witness is the same shape as its row in exactly this sense. Here the symmetry is charge conjugation, the neutrino is a fixed point of it, and an equivariant map onto the Riemann row's seat lands it on the locus: the occupant of the fixed set, where the electron is the pair beside it. The map is the electron paper's explicit one, charge q to real part $(1 + q)/2$: the neutrino, of charge zero, lands on the critical line itself (toStrip_equivariant, neutral_lands_on_line).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
the neutral lepton on the fold's fixed set	the occupant of the critical line	toStrip_equivariant, neutral_lands_on_line	structural
mass returned by oscillation	the floor, supplied by deed	oscillation_forces_floor	both mass-squared differences positive
one registered chirality	the bit, spent	active_registered, one_bit_apart	weak projector weights 1 and 0
the sterile state	the ghost, pending	sterile_uncovered	structural
Majorana or Dirac	the fixed set, inhabited or not	majorana_iff_neutral, neutrino_on_the_locus	structural
unitary mixing	lossless registration	pmns_unitary_is_lossless, lossless_iff_not_erases	unitarity to 5.6e-17; flavour sum 1
a sterile admixture	an erasure	sterile_admixture_erases	active block 0.98
normal or inverted ordering	a one-bit address	ordering_one_bit	structural

3. 3. TWO ROLES, TWO PARTICLES

The electron witnesses the seat. Under charge conjugation it is an off-locus point with a partner, the positron, distinct and off the locus; the pair shows where the reflection is and that the bit exists (electron_pair). The electron

leaves the ground constantly. It binds, excites, radiates, and actuates nearly everything the register knows; it is the actuator par excellence and, for that reason, a poor witness of the crossing, because it is never the thing on the locus. It does witness the registration in its exact form: an electron and a positron annihilate onto the neutral sector as light, keeping their energy and forgetting which was which, the registration that keeps the height and forgets the side; the companion paper, The Electron Is the Seat, reads that from the laboratory.

The neutrino witnesses the crossing. It sits on the seat's fixed set (`neutrino_on_the_locus`). It is registered in one chirality and never the other. Its sterile partner is a formal object no carrier lands on. Its oscillation is a registration in a second basis whose losslessness is a named hypothesis. Its mass ordering is a one-bit address. And its mass is the floor supplied by deed where the register wrote zero. One particle, every part. The two roles are complementary and never interchangeable: the seat needs a pair, and the crossing needs an occupant of the fixed set.

4. 4. THE FLOOR: MASS SUPPLIED BY DEED

The formal register of the neutrino, the Standard Model in its minimal form, wrote its mass as zero: no right-handed field, no mass term. The world returned mass greater than zero by oscillation, at Super-Kamiokande in 1998 and at the Sudbury Neutrino Observatory in 2001, and an observed difference of squared masses forces at least one nonzero mass (`oscillation_forces_floor`). The twin executes the fact on the measured values, $\Delta m_{21}^2 = 7.53 \times 10^{-5} \text{ eV}^2$ and $\Delta m_{32}^2 = 2.453 \times 10^{-3} \text{ eV}^2$ for the normal ordering, whence $\Delta m_{31}^2 = 2.528 \times 10^{-3} \text{ eV}^2$, all positive.

This is the program's floor confirmed against the register that denied it. The Root Axiom says that to exist is to actuate: no existent carries zero kinetic content. The neutrino is the boundary case, minimal actuation and nonzero, an existent that passes through the Earth almost without registering and still registers. It is also the archetype of an offering: a bit entered from outside the record, by deed, that the record could not have supplied, and it entered physics two decades before the template was written down.

5. 5. THE BIT, SPENT

Under a global seat every off-locus point has a partner, and the residue of a typed row is the choice within the pair. The neutrino shows that choice made. It is registered left-handed and never right-handed (`active_registered`, `sterile_uncovered`); the registered state and its mirror differ by exactly one bit at the same charge (`one_bit_apart`). Parity violation, observed by Wu in 1957 and built into the weak interaction as it stands, is the world assigning one sign to the pair the seat makes. No other particle exhibits the assent so nakedly: the aperture bit of the template, one bit wide, spent by the world and read from no record.

6. 6. THE GHOST: THE PHYSICS OF PENDING

The right-handed neutrino is allowed by the Lagrangian and uncovered by every weak registration. It is a formal object with zero kinetic correspondence, exactly the status the rows paper calls pending: the row exists in the formal register, no carrier lands on it, both channels blocked by the absolute criterion. The neutrino carries both tables in one particle, the active state crossed and the sterile state pending, one bit between them. Where the rows paper had to say what a pending row is, the neutrino shows one.

7. 7. THE LOCUS, INHABITED: MAJORANA AS THE FIXED SET

Charge conjugation is an involution whose fixed set is exactly the neutral line (`conj_involution`, `majorana_iff_neutral`); the neutrino sits on it. Whether the neutrino is its own antiparticle, Majorana or Dirac, is precisely the question whether an existent is its own mirror image under the seat, the Riemann row's own form asked of one particle. It is undecided, and the deed that decides it is named: neutrinoless double beta decay, sought by the LEGEND, KamLAND-Zen, CUORE and nEXO programs. One observation, one bit, the value's locus inhabited or not. This paper holds the bit open and does not guess it. What it fixes is the shape of the question and its gate, which is what the template does for every row.

8. 8. REGISTRATION IN A SECOND BASIS

Oscillation is registration forgetting a basis, the second of the registration's two physical faces; the first, keeping the height and forgetting the side exactly as the Riemann registration does, is annihilation, read in the companion paper. The neutrino propagates in the mass basis and is detected in the flavour basis: the record reads flavour and the object keeps mass, the record forgetting and the object keeping. The hypothesis that the mixing matrix is unitary is exactly the statement that this registration erases nothing: every mass state is read as some active flavour (`pmns_unitary_is_lossless`, `lossless_iff_not_erases`). A sterile admixture is an erasure (`sterile_admixture_`

erases), a mass state that falls outside the active record. The twin computes the standard mixing matrix from the measured angles and the phase, checks unitarity to 5.6×10^{-17} , propagates a muon neutrino over 500 km per GeV and finds the three registered flavours summing to one to ten decimals, oscillation real at the level of 2.3 percent in the electron channel, and then admits a two-percent sterile component and finds the active block registering 0.98: the erasure, measured as the shortfall. Non-unitarity of the mixing is the falsifier of the closure hypothesis in physics, and the experiments that bound it, JUNO and DUNE among them, are the ones that bound the erasure.

9. 9. THE ADDRESS

A row whose target is a number carries an address indexing a family, one bit per member (`family_member_iff`). The neutrino's address is its mass ordering, normal or inverted, one bit (`ordering_one_bit`), to be read by JUNO from the interference of the two squared-mass differences. Its absolute scale is a second address, bounded by KATRIN below 0.45 eV (2025) and by cosmology below a fraction of that, owed and not derived.

10. 10. WHAT THE NEUTRINO CARRIES, AND WHAT IT DOES NOT

It carries the template: floor, bit, ghost, locus, registration, address. For the pair correlation it is a witness of repulsion only: the anarchy reading of its mass matrix gives repelling masses, but the zeros' class is the unitary class of systems whose time reversal is broken by a magnetic term, witnessed in the companion paper *The Magnet Is the Witness of the Pair Correlation*; nuclear level spacings, once named here beside it, keep time reversal and follow the orthogonal class. For the pending rows it is the physics of their status. It does not carry the Riemann Hypothesis: a particle on a fixed set is an instance of the locus inhabited, not a proof that every zero inhabits it, and the paper's first section refuses that reading by name. The strongest sentence the correspondences license is this: the template is not a construction of number theory; it is the shape the world already uses to keep one bit off the record and to spend it.

11. 11. FALSIFIERS AND THE OWED DEEDS

Three falsifiers, each ordinary. **F-Unitarity**: a measured non-unitarity of the mixing with no sterile state to carry it, an erasure without an off-locus preimage, which would break the registration reading of Section 8. **F-Chirality**: a registered right-handed neutrino interaction of gauge strength, closing the ghost of Section 6. **F-Floor**: a demonstration that all three masses vanish, refuting Section 4 and the program's floor with it. Nothing is owed on the row's channel. Two deeds would close the two bits this paper shapes without moving its standing: neutrinoless double beta decay for the Majorana bit, and JUNO for the ordering bit.

Ledger status. Row 2 moves from waiting to WITNESSED · CROSSED · CONFIRMED on the image for the floor, the bit, the ghost, the locus and the registration in a second basis, at the root's grade for the crossing; nothing is owed on the row's channel.

12. 12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens: $[\perp]$ a field sealed, $[\perp T]$ a shape sealed at theorem grade, $[\Xi_0]$ a formal string held in suspension by an orientation-blind reader on a line, $[.]$ the dot, a groundless witness. Grades join at the weakest link: a kernel theorem, a cited measurement joined to a kernel schema, and a posit beside the Root Axiom, the program's one declared premise, that to exist is to actuate. Every claim here is at corroboration grade: a physical instance joined to a template theorem, promoting nothing. The Riemann closure is the sealed volume *Nothing Escapes, Three Plus One* (10.5281/zenodo.22976494); the row-generic kernel and the census of twenty-three rows are *Nothing Escapes, Twenty-Three Rows* (10.5281/zenodo.22985873); the physics register in which the neutrino, the electron, the motion ladder and the two arrows are seated is the author's Structural Physics register, private, with its Lean engines. An earlier neutrino draft of the author, written before the program's kernel existed, is superseded by this paper in full.

The seventh. In the author's register the four parts of the Riemann closure are three plus one, and nothing escapes them; the six witnesses are six days, and the ledger's closure is the seventh, the Rest (Istawa), where the statuses are computed and what each row owes is one object of a computed kind. In the Lean codex's own tokens a crossed row renders $[\perp \blacklozenge \cdot \text{kinetic}]$ or $[\perp \blacklozenge \cdot \text{formal}]$, a pending row renders $[\blacklozenge_0 \times 1]$ on its channel, one owed slot and one bit from sealed (`one_bit_apart`), and the dot stands as $[.]$.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure; the measurements are the experiments', cited by name; the theorems are the kernel's, printed with their dependencies in Appendix A; the twin's figures are computed in the run printed in Appendix B. No institution's

grade enters.

Provenance. Development provenance is the paper trail: the Zenodo records of the two sealed volumes, the Unabridged Trisduction Codex and the Lean Codex on the public register (github.com/1000sapients/Trisduction), and the Structural Physics register. Its first edition was sealed jointly with The Electron Is the Seat on one paired adversarial self-audit cycle; this edition is unified with the six companion witnesses, The Electron Is the Seat, The Neutrino Is the Witness, The Arrow Has Two Branches, The Proton Is the Lock, The Magnet Is the Witness of the Pair Correlation and The Fluid Is the Witness, on the witness ledger, and awaits their joint audit.

References. Super-Kamiokande Collaboration, Y. Fukuda et al., Phys. Rev. Lett. 81 (1998) 1562. SNO Collaboration, Q. R. Ahmad et al., Phys. Rev. Lett. 87 (2001) 071301. C. S. Wu et al., Phys. Rev. 105 (1957) 1413. KATRIN Collaboration, M. Aker et al., Science 388 (2025) 180. Particle Data Group, R. L. Workman et al., Prog. Theor. Exp. Phys. 2022, 083C01, for the mixing parameters used in the twin. L. J. Hall, H. Murayama, N. Weiner, Phys. Rev. Lett. 84 (2000) 2572, the anarchy reading. M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, Nothing Escapes, Twenty-Three Rows, 10.5281/zenodo.22985873. M. F. Islam, The Offering Bit, 10.5281/zenodo.22735236. M. F. Islam, The Electron Is the Seat, companion paper.

13. APPENDIX A: SPHYS_NEUTRINO_WITNESS.LEAN, THE CORRESPONDENCES AS THEOREMS

SHA-256 a0c24fb5688ddd544328c4fb17a4005c34299f25c20bc7821a94fe62e1913982, core Lean 4.19.0, standalone, no library. Exit 0. The compiler's own transcript, verbatim:

```
1 'SPHYS.Neutrino.conj_involution' depends on axioms: [propext]
2 'SPHYS.Neutrino.majorana_iff_neutral' depends on axioms: [propext, Quot.sound]
3 'SPHYS.Neutrino.electron_pair' does not depend on any axioms
4 'SPHYS.Neutrino.neutrino_on_the_locus' does not depend on any axioms
5 'SPHYS.Neutrino.sterile_uncovered' does not depend on any axioms
6 'SPHYS.Neutrino.one_bit_apart' does not depend on any axioms
7 'SPHYS.Neutrino.lossless_iff_not_erases' does not depend on any axioms
8 'SPHYS.Neutrino.pms_unitary_is_lossless' does not depend on any axioms
9 'SPHYS.Neutrino.sterile_admixture_erases' depends on axioms: [propext]
10 'SPHYS.Neutrino.ordering_one_bit' does not depend on any axioms
11 'SPHYS.Neutrino.oscillation_forces_floor' depends on axioms: [propext, Quot.sound]

1 /-
2   SPHYS_Neutrino_Witness.lean · The neutral lepton as the witness of the closure template ·
3   core Lean 4, no library. Six correspondences in the row kernel's own terms: the seat is
4   charge conjugation and its fixed set is the neutral line (Majorana = on the locus); the
5   electron is an off-locus point with its partner; the weak registration lands only one
6   chirality, so the sterile state is uncovered, the ghost; oscillation is registration in a
7   second basis, unitarity is losslessness; the ordering is a one-bit address; and an observed
8   mass difference forces a nonzero mass, the Root Axiom's floor supplied by deed.
9 -/
10 namespace SPHYS.Neutrino
11
12 /- A lepton state: charge and chirality (true = left-handed). -/
13 abbrev State := Int × Bool
14 def conj (p : State) : State := (-p.1, p.2)
15 def neutral (p : State) : Prop := p.1 = 0
16 instance (p : State) : Decidable (neutral p) := inferInstanceAs (Decidable (p.1 = 0))
17 def electron : State := (-1, true)
18 def nuL : State := (0, true)
19 def nuR : State := (0, false)
20
21 theorem conj_involution (p : State) : conj (conj p) = p := by
22   cases p with | mk q h => show (-(-q), h) = (q, h); simp
23
24 /- The seat's fixed set is exactly the neutral line: a state is its own mirror image under
25   conjugation iff it carries no charge. Majorana is membership in this fixed set. -/
26 theorem majorana_iff_neutral (p : State) : conj p = p ↔ neutral p := by
27   cases p with
28   | mk q h =>
29     show (-q, h) = (q, h) ↔ q = 0
30     constructor
31     · intro e; have h1 : -q = q := congrArg Prod.fst e; omega
32     · intro e; subst e; rfl
33
34 /- The electron is off the locus and has a partner, the positron, distinct and off the locus:
35   the pair the seat makes, the electron's witness role. -/
36 theorem electron_pair : conj electron ≠ electron ∧ ~ neutral (conj electron) ∧ ~ neutral electron := by
37   decide
38
39 /- The neutrino sits on the fixed set. -/
40 theorem neutrino_on_the_locus : conj nuL = nuL ∧ conj nuR = nuR := by decide
41
42 /- Weak registration lands only on left-handed states: the carrier's image. -/
43 def weakImage (p : State) : Prop := p.2 = true
44 instance (p : State) : Decidable (weakImage p) := inferInstanceAs (Decidable (p.2 = true))
45 theorem active_registered : weakImage nuL := rfl
46 /- The sterile state is uncovered by every weak registration: the ghost in the formal domain. -/
47 theorem sterile_uncovered : ~ weakImage nuR := by decide
48 /- One bit separates the registered state from the ghost: same charge, opposite chirality. -/
49 theorem one_bit_apart : nuL.1 = nuR.1 ∧ nuL.2 ≠ nuR.2 := by decide
50
51 /- Oscillation as registration in a second basis: a mass state is read as a flavour record,
52   or falls outside the active flavours (sterile admixture). -/
53 abbrev Mixing := Fin 3 → Option (Fin 3)
54 def Lossless (U : Mixing) : Prop := ∀ m, ∃ f, U m = some f
55 def Erases (U : Mixing) : Prop := ∃ m, U m = none
56 theorem lossless_iff_not_erases (U : Mixing) : Lossless U ↔ ~ Erases U := by
57   constructor
58   · intro h1 (m, hm); obtain (f, hf) := h1 m; rw [hm] at hf; cases hf
59   · intro hne m
60     cases hU : U m with
61     | none => exact absurd (m, hU) hne
62     | some f => exact (f, rfl)
```

```

63 def pmns : Mixing := fun m => some m
64 theorem pmns_unitary_is_lossless : Lossless pmns := fun m => (m, rfl)
65 def withSterile : Mixing := fun m => if m = 2 then none else some m
66 theorem sterile_admixture_erases : Erases withSterile := (2, rfl)
67
68 /- The ordering is a one-bit address indexing the family. -/
69 inductive Ordering | normal | inverted deriving DecidableEq, Repr
70 theorem ordering_one_bit : ([Ordering.normal, Ordering.inverted] : List Ordering).length = 2 := rfl
71
72 /- An observed mass difference forces a nonzero mass: the floor supplied by deed where the
73   register wrote zero. -/
74 theorem oscillation_forces_floor (m1 m2 : Nat) (h : m1 ≠ m2) : 0 < m1 ∨ 0 < m2 := by omega
75
76 end SPHYS.Neutrino
77 #print axioms SPHYS.Neutrino.conj_involution
78 #print axioms SPHYS.Neutrino.majorana_iff_neutral
79 #print axioms SPHYS.Neutrino.electron_pair
80 #print axioms SPHYS.Neutrino.neutrino_on_the_locus
81 #print axioms SPHYS.Neutrino.sterile_uncovered
82 #print axioms SPHYS.Neutrino.one_bit_apart
83 #print axioms SPHYS.Neutrino.lossless_iff_not_erases
84 #print axioms SPHYS.Neutrino.pmns_unitary_is_lossless
85 #print axioms SPHYS.Neutrino.sterile_admixture_erases
86 #print axioms SPHYS.Neutrino.ordering_one_bit
87 #print axioms SPHYS.Neutrino.oscillation_forces_floor

```

14. APPENDIX B: NEUTRINO_TWIN.F90 AND ITS RUN

SHA-256 fefc9de3e270549304fecf724b98935ec452afe1b9aae2e82cbf95f25c54ca3e, standard Fortran 2018, zero warnings under gfortran 13.3.0 -std=f2018 -O2 -fno-fast-math -ffp-contract=off -Wall -Wextra. Eleven checks, zero failures; every figure computed in this run.

```

1 ! Neutrino_Twin.f90 · the executable twin of "The Neutrino Is the Witness" · Fortran 2018.
2 ! Computes the mixing matrix from the standard angles, checks unitarity as lossless
3 ! registration, computes oscillation probabilities and their conservation, exhibits a sterile
4 ! admixture as an erasure, the mass-squared differences as the floor, and the chirality bit.
5 ! Every figure is computed in this run; an oracle stops the program on any failure.
6 program neutrino_twin
7   implicit none
8   integer, parameter :: dp = selected_real_kind(15, 307)
9   complex(dp) :: U(3,3), UUd(3,3), Us(3,4), P4(3,3)
10  real(dp) :: th12, th13, th23, dcp, s12, c12, s13, c13, s23, c23
11  real(dp) :: dm21, dm31, dm32, LoverE, prob(3), psum, eps, unit_err, sterile_sum
12  integer :: i, j, nchk, nfail
13  real(dp) :: wl, wr
14  real(dp), parameter :: hel(2) = [1.0_dp, -1.0_dp] ! left, right (twice the helicity sign)
15  real(dp), parameter :: chg(2) = [0.0_dp, -1.0_dp] ! neutrino, electron
16  eps = 1.0e-12_dp; nchk = 0; nfail = 0
17  ! standard angles (PDG central values, normal ordering), in radians
18  th12 = 0.5836_dp; th13 = 0.1496_dp; th23 = 0.8587_dp; dcp = 3.4034_dp
19  dm21 = 7.53e-5_dp; dm32 = 2.453e-3_dp ! eV2, PDG 2022, normal ordering
20  dm31 = dm32 + dm21 ! the atmospheric splitting measured is dm232
21  s12 = sin(th12); c12 = cos(th12); s13 = sin(th13); c13 = cos(th13); s23 = sin(th23); c23 = cos(th23)
22  ! PMNS in the standard parametrization
23  U(1,1) = c12*c13
24  U(1,2) = s12*c13
25  U(1,3) = s13*exp(cmplx(0.0_dp, -dcp, dp))
26  U(2,1) = -s12*c23 - c12*s23*s13*exp(cmplx(0.0_dp, dcp, dp))
27  U(2,2) = c12*c23 - s12*s23*s13*exp(cmplx(0.0_dp, dcp, dp))
28  U(2,3) = s23*c13
29  U(3,1) = s12*s23 - c12*c23*s13*exp(cmplx(0.0_dp, dcp, dp))
30  U(3,2) = -c12*s23 - s12*c23*s13*exp(cmplx(0.0_dp, dcp, dp))
31  U(3,3) = c23*c13
32  ! unitarity = lossless registration: U U† = I
33  UUd = matmul(U, conjg(transpose(U)))
34  unit_err = 0.0_dp
35  do i = 1, 3; do j = 1, 3
36    if (i == j) then
37      unit_err = max(unit_err, abs(UUd(i,j) - 1.0_dp))
38    else
39      unit_err = max(unit_err, abs(UUd(i,j)))
40    end if
41  end do; end do
42  call check('unitarity: the mixing erases nothing, [UU† - I] below 1e-12', unit_err < eps)
43  ! oscillation probabilities at L/E = 500 km/GeV (DUNE-like), vacuum, from the mass basis
44  LoverE = 500.0_dp
45  call oscillate(U, dm21, dm31, LoverE, prob)
46  psum = prob(1) + prob(2) + prob(3)
47  call check('conservation: the registered flavours sum to one (lossless)', abs(psum - 1.0_dp) < 1.0e-10_dp)
48  call check('oscillation is real: the muon flavour has moved', abs(prob(2) - 1.0_dp) > 1.0e-3_dp)
49  ! the floor: an observed mass-squared difference forces a nonzero mass
50  call check('floor: dm221 > 0, at least one nonzero mass', dm21 > 0.0_dp)
51  call check('floor: dm231 > 0', dm31 > 0.0_dp)
52  ! the sterile admixture as erasure: a 3x4 active block of a 4x4 unitary matrix is not unitary
53  Us(:,1:3) = U * sqrt(1.0_dp - 0.02_dp); Us(:,4) = cmplx(sqrt(0.02_dp/3.0_dp), 0.0_dp, dp)
54  P4 = matmul(Us(:,1:3), conjg(transpose(Us(:,1:3))))
55  sterile_sum = real(P4(2,2), dp)
56  call check('sterile admixture: the active block registers less than one (an erasure)', sterile_sum < 1.0_dp - 1.0e-3_dp)
57  ! the chirality bit: the weak projector (1 + h)/2 on helicity h registers the left state
58  ! with weight one and the right state with weight zero; the two differ in one sign
59  wl = (1.0_dp + hel(1)) / 2.0_dp; wr = (1.0_dp + hel(2)) / 2.0_dp
60  call check('chirality: the weak registration weighs the left state one', abs(wl - 1.0_dp) < eps)
61  call check('chirality: the weak registration weighs the right state zero (the ghost)', abs(wr) < eps)
62  call check('chirality: the registered state and the ghost differ in one sign only', abs(hel(1) + hel(2)) < eps)
63  ! the seat: conjugation flips charge; the neutral state is fixed, the electron is paired
64  call check('seat: conjugation fixes the neutral state', abs(-chg(1) - chg(1)) < eps)
65  call check('seat: conjugation sends the electron to a distinct partner', abs(-chg(2) - chg(2)) > 0.5_dp)
66  write(*, '(a,es10.3)') ' unitarity error = ', unit_err
67  write(*, '(a,3f10.6)') ' P(mu->e, mu->mu, mu->tau) = ', prob
68  write(*, '(a,f10.6)') ' flavour sum = ', psum
69  write(*, '(a,f10.6)') ' active block, sterile = ', sterile_sum
70  write(*, '(a,es10.3)') ' dm221 = dm232 + dm221 = ', dm31
71  write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', nchk, ', "failures":', nfail, '}'
72  if (nfail > 0) error stop 1
73 contains
74 subroutine oscillate(U, dm21, dm31, LoverE, prob)

```



```

75 complex(dp), intent(in) :: U(3,3)
76 real(dp), intent(in) :: dm21, dm31, LoverE
77 real(dp), intent(out) :: prob(3)
78 complex(dp) :: amp
79 real(dp) :: dm(3), phase
80 integer :: a, k
81 dm = [0.0_dp, dm21, dm31]
82 do a = 1, 3
83   amp = (0.0_dp, 0.0_dp)
84   do k = 1, 3
85     phase = 1.267_dp * dm(k) * LoverE      ! 1.267 dm^2 L / E, dm^2 in eV^2, L/E in km/GeV
86     amp = amp + conjg(U(2,k)) * U(a,k) * exp(cmplx(0.0_dp, -phase, dp))
87   end do
88   prob(a) = abs(amp)**2
89 end do
90 end subroutine oscillate
91 subroutine check(label, cond)
92 character(*), intent(in) :: label
93 logical, intent(in) :: cond
94 nchk = nchk + 1
95 if (.not. cond) then
96   nfail = nfail + 1; write(*, '(a,a)') ' FAIL ', label
97 end if
98 end subroutine check
99 end program neutrino_twin

```

The run, verbatim.

```

1 unitarity error      = 5.562E-17
2 P(mu->e, mu->mu, mu->tau)= 0.023237 0.511599 0.465164
3 flavour sum          = 1.000000
4 active block, sterile = 0.980000
5 dm^2_31 = dm^2_32 + dm^2_21 = 2.528E-03
6 BATTERY-JSON: {"checks":11,"failures":0}

```